Amteshwar Raj Singh

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EDUCATION

B.Tech CSE (AI & ML) — Dev Bhoomi Uttarakhand University (2024–2028)
Coursework: Machine Learning, Deep Learning, Computer Vision, Data Structures, Probability & Statistics

PROJECTS & RESEARCH

SteelVision — CNN-Based Steel Defect Detection (Streamlit + Research Paper)

- Built CNN classifier on ~70k steel images achieving ~94% accuracy under real-world variations
- Developed Streamlit real-time inspection system for prediction and visualization
- Authored research paper detailing experiments, dataset design, and evaluation (GitHub preprint)

Road Crack Detection — YOLOv8 Object Detection

- Trained YOLOv8 model on ~55k annotated images achieving mAP@0.5 ≈ 0.92 with stable inference
- Improved robustness using augmentation, validation analysis, and error inspection
- Implemented real-time detection pipeline using PyTorch and OpenCV

DiabetesGuard-AI — Predictive Healthcare ML System

- Built interpretable diabetes risk prediction model using structured medical features
- Performed preprocessing, feature analysis, and visualization for explainable predictions

Suraksha AI — Real-Time Public Safety Monitoring (YOLO + Pathway + Computer Vision)

- Built an AI-based safety monitoring system for real-time threat detection using computer vision models
- Implemented YOLO-based detection pipeline for identifying suspicious activities and safety violations from video streams
- Integrated real-time event processing using Pathway to analyze continuous data streams and generate alerts
- Designed automated monitoring workflow for public places with live detection, event logging, and alert system

ACHIEVEMENTS

- Finalist — IIT Delhi BECon 2026
- Finalist— Microsoft Green Hack for Bharat (Out of 7500 teams got 13th rank)
- Finalist — Multiple university AI/ML hackathons

LEADERSHIP

President, Robotics & Gaming Society (RGS)

- Led robotics and AI initiatives and mentored student ML project teams
- Organized inter-university drone seminar with 100+ participants

Technical Skills

Programming: Python (Libraries: NumPy, SciPy, Matplotlib, OpenCV , Pandas , Numpy, Computer Vision, Industrial AI)
ML/CV: CNNs, YOLOv8, Object Detection, Model Evaluation
Frameworks: PyTorch, TensorFlow
Tools: Git, Jupyter, Streamlit